COMPLEX ENGINEERING PROBLEM

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CEE308: Reinforced Concrete Design I

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Abstract

This report presents the comprehensive structural design of a three-story reinforced concrete school building situated on a 120' × 180' plot, intended for educational use as per the occupancy requirements outlined in ACI 318-14 and ASCE 7. The structure adopts a reinforced concrete frame system, and the design scope includes key structural elements such as a one-way slab, singly reinforced beam, doubly reinforced beam, an exterior square column, and an interior circular spiral column. The design approach utilizes manual calculations based on fundamental reinforced concrete design formulas along with design aids and relevant ACI code tables. The grid layout was selected to satisfy the criteria for a one-way slab system (LY/LX > 2), ensuring efficient load transfer and simplified analysis. Each structural member is designed for strength, stability, and serviceability, including checks for deflection control as specified in ACI Tables 7.3.1.1 and 9.3.1.1. Live loads were assumed according to the intended function of the building, with classroom occupancy set at 40 psf, and appropriate material strengths of **fc' = 3 ksi** for concrete and **fy = 60 ksi** for steel were used throughout. The reinforcement detail covering bar sizes, spacing, development lengths, and shear reinforcement—was prepared in accordance with ACI requirements and presented through structural drawings developed in AutoCAD/Revit on A4 sheets. This report documents the design process and calculations while reflecting on the rationale behind each decision, demonstrating the practical application of code-based design for educational infrastructure.

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1. Introduction

This report presents the structural design of a three-story reinforced concrete school building on a 120' × 180' plot, following the occupancy requirements of **ACI 318-14** and **ASCE 7**. The building is designed as a reinforced concrete frame system and includes essential structural elements such as a one-way slab, singly reinforced beam, doubly reinforced beam, an exterior square column, and an interior circular spiral column. Manual calculations were performed using standard reinforced concrete design formulas, supported by ACI code tables and design aids. A **15' × 30'** grid layout was selected to meet the condition **LY/LX > 2** for one-way slab design, allowing efficient load transfer. Each structural element was checked for strength and serviceability, including deflection control based on ACI Tables **7.3.1.1** and **9.3.1.1.** The assumed live load for classrooms was 40 psf, and material strengths of fc' = 3 ksi for concrete and fy = 60 ksi for steel were used. Reinforcement detailing, including bar sizes, spacing, development lengths, and stirrup-placement, was carried out in accordance with ACI standards. Structural drawings for all components were prepared using AutoCAD/Revit on A3 sheets. This report highlights the complete design process and justifies design decisions, ensuring code compliance, structural safety, and serviceability in a functional educational building.

2. Literature Review

The structural design of educational buildings, particularly reinforced concrete (RC) school buildings, must ensure safety, durability, and serviceability in compliance with modern design codes such as **ACI 318-14** and **ASCE 7**. Numerous studies have explored the optimization of structural systems using reinforced concrete frames for medium-rise buildings, emphasizing load transfer, material efficiency, and code compliance **(Mehta & Monteiro, 2014).**

One-way slab systems are often preferred in school buildings due to their simplicity and ease of construction. According to **Nilson et al. (2010)**, LY/LX > 2 is a practical criterion that ensures slabs behave structurally as one-way members. These slabs, when supported

adequately, provide efficient load transfer while maintaining serviceability through controlled deflection, as mandated by **ACI Tables 7.3.1.1 and 9.3.1.1**.

The use of singly and doubly reinforced concrete beams is essential for balancing flexural and shear demands. The need for compression reinforcement in doubly reinforced sections arises when space constraints or moment requirements exceed the capacity of singly reinforced members (**MacGregor & Wight, 2012**). Research by **Mirmiran et al. (2002)** has shown the importance of ductility and bar anchorage for beams in educational facilities to withstand dynamic and live loads effectively.

The columns, particularly spirally reinforced circular columns, are critical in resisting axial and moment loads, especially in seismic-prone areas. Studies highlight that spiral reinforcement enhances ductility and confinement (**Park & Paulay, 1975**). The **ACI 318-14** provisions ensure these members meet both strength and deformation requirements under eccentric loading, which is a common design scenario in school infrastructure.

Designing using manual methods, as emphasized in engineering education, provides a deeper understanding of behavior and reinforces code interpretation skills. While software tools expedite analysis, manual calculations grounded in ACI 318-14 formulas and tables help in verifying automated results and understanding design assumptions.

Finally, detailing practices, including bar development length, clear cover, and spacing per ACI standards, are paramount to durability and corrosion resistance, especially in long-span elements like slabs and beams exposed to heavy use in school buildings.

3. Methodology

3.1 Design Philosophy

The structural design was carried out in accordance with **ACI 318-14**, with load requirements referenced from **ASCE 7** for buildings intended for educational occupancy. The design approach primarily involved manual calculations using standard reinforced concrete

design formulas. Design aids, particularly Table A.8 from the ACI Design Handbook, were utilized to ensure compliance with deflection criteria. This methodology ensured that the structural elements met both strength and serviceability requirements, while adhering to code-specified safety factors and material limitations.

3.2 Grid Layout Selection

A grid layout of 15’ by 30’ was adopted for the building to facilitate the design of one-way slabs. This layout was intentionally selected to satisfy the condition of **LY/LX > 2,** which is a necessary criterion for one-way slab behavior according to ACI recommendations. The chosen spacing provided structural efficiency and allowed for straightforward distribution while optimizing the use of materials and formwork during construction.

3.3 Load Assumptions

The loads used in the design were based on occupancy as a school building. A live load of 40 pounds per square foot (psf) was adopted for classrooms, following the specifications provided in ASCE 7. Dead loads were calculated based on the self-weight of structural elements and the unit weight of materials typically used in reinforced concrete construction. The characteristic compressive strength of concrete was assumed as $\mathbf{f_c' = 3\ ksi}$, while the yield strength of steel reinforcement was taken as $\mathbf{f_y = 60\ ksi}$. All beams were reinforced with two-legged stirrups to resist shear forces, and detailing followed ACI guidelines for development length, spacing, and cover.

4. Hand Calculations

f'_c = 3000 psi , f_y = 60000 psi

class room = 40 lb/ft²

SLAB:-

$$\frac{L_y}{L_x} = \frac{30}{15} = 2$$

It will be design as one way slab.

$$h = \frac{\ell}{24} = \frac{15 \times 12}{24} = 7.5''$$

$h = 8''$

$d = 7''$

$$\Rightarrow D.L = \frac{8 \times 150}{12} = 100 \text{ lb/ft}^2$$

$$M_u = \frac{w_u l^2}{8}$$

$$W_u = 1.2\,DL + 1.6\,LL$$
$$= 1.2(100) + 1.6(40)$$
$$W_u = 184 \text{ lb/ft}^2$$

$$M_u = \frac{0.184 \times 15^2}{8}, \quad M_u = 5.175 \text{ k-ft}$$

$$M_n = Rbd^2, \quad \frac{M_u}{\phi} = Rbd^2$$

$$K = \frac{M_u}{\phi bd^2} = \frac{5.175}{0.9 \times 12 \times 7^2} = 117 \text{ psi}$$

$\rho = 0.0020$

$$A_{s\,req} = \rho_{req} \times b \times d = 0.0020 \times 12 \times 7 = 0.168 = 0.18 \text{ in}^2/\text{ft}$$

#3 @ 8" c/c (Table A3)

Must be less than 3h or 18"

$b = 12''$, $d = 22''$, $h = 24''$

$DL = \frac{12}{12} \times \frac{24}{12} \times 150 \Rightarrow DL = 300\ lb/ft$

$W_u = 1.2(300 + 750) + 1.6(300)$

$W_u = 1.740\ k/ft$

$M_u = \frac{W_u l^2}{8} = \frac{1.74 \times 30^2}{8} = 196\ k\cdot ft$

$R = \frac{M_u}{\phi bd^2} = \frac{196 \times 12000}{0.9 \times 12 \times 22^2} = 450\ psi$

$f = 0.0084$ from table A-5-A

$f_{0.005} = 0.0135$, $f_{0.004} = 0.0155$

$f_{act} < f_{0.005} < f_{0.004}$

$\phi = 0.9$ (underreinforced)

$A_s = fbd = 0.0084 \times 12 \times 22 = 2.22\ in^2$

3#8, $A_s = 2.37\ in^2$

$f = \frac{A_s}{bd} = \frac{2.37}{12 \times 22} = 0.0089$

$f_{act} < f_{0.005} < f_{0.004}$

3#8

→ Table A-7

Satisfy max no bar in one layer

→ Table A-8

3#8 also satisfy minimum no of Grade

So bar as a single layer.

$a = \frac{A_s f_y}{0.85 f'_c b} = \frac{2.37 \times 60{,}000}{0.85 \times 3000 \times 12}$

$a = 4.65''$

Smax

$$Mn = As\,fy\left(d - \frac{a}{2}\right) = 2.37 \times 60{,}000\left(22 - 4.65/2\right)$$

$Mn = 234$ k·ft

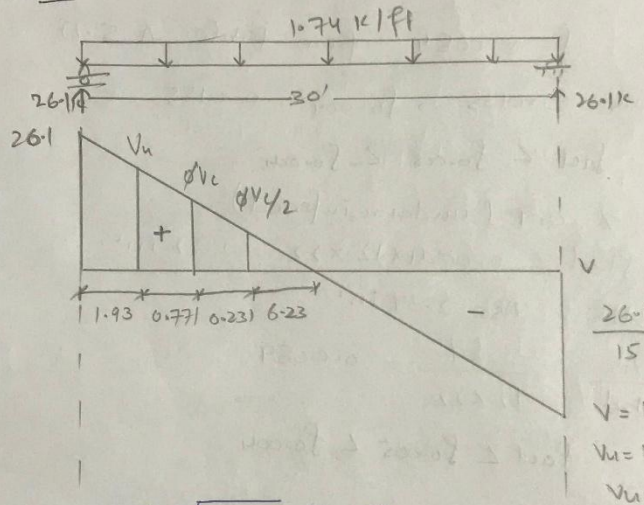
$\phi Mn = 210$ k·ft

$\phi Mn > Mu$

$$ld = \left(\frac{3}{40} \times \frac{60{,}000}{1\times\sqrt{3000}} \times \frac{1\times1\times1}{\left(\frac{2}{1}+1\right)}\right) \times 1$$

$ld = 41''$

SHEAR DESIGN:



$$\frac{26.1}{15} = \frac{V}{(15-x)}$$

$V = 1.74(15-x)$

$Vu = 1.74(15-1.83)$

$Vu = 23$ kip

- $\phi Vc = 10.75 \times 2\sqrt{3000} \times 12 \times 22$

$\phi Vc = 21.7$ kip

$\phi Vc/2 = 10.85$ kip , $\phi Vs = Vu - \phi Vc = 1.73$

$Vu > \phi Vc$

web reinforcement required

$Vs < 2Vc$ (smax)

- $$Sd = \frac{\phi Av\,fyt\,d}{Vu - \phi Vc} = \frac{0.75(0.22)(60{,}000)(22)}{23 - 21.7}$$

$Sd = 161''$

$\rightarrow Mu = 0.85(3000)(5.1)(12)\left(\frac{12}{2} - \frac{5.1}{2}\right) + \left(\frac{3.6}{2}\right)(58000 - 0.85(3000))\left(\frac{12}{2} - 2\right)$

$+ \left(\frac{3.6}{2}\right)(58000)\left(10 - \frac{12}{2}\right)$

$Mu = \frac{538407 + 399240 + 417600}{12000}$

$Mu = 113 K\text{-}ft$

$\phi Pn = 0.75 \times 151 = 113 kip$

$\phi Mn = 0.75 \times 113 = 84 K\text{-}ft$

$\rho_{min} < \rho < \rho_{max}$

$\phi Pn > Pu$

$\phi Mn > Mu$] ok

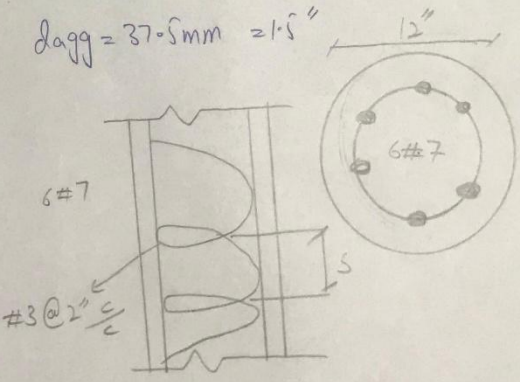
Ties:

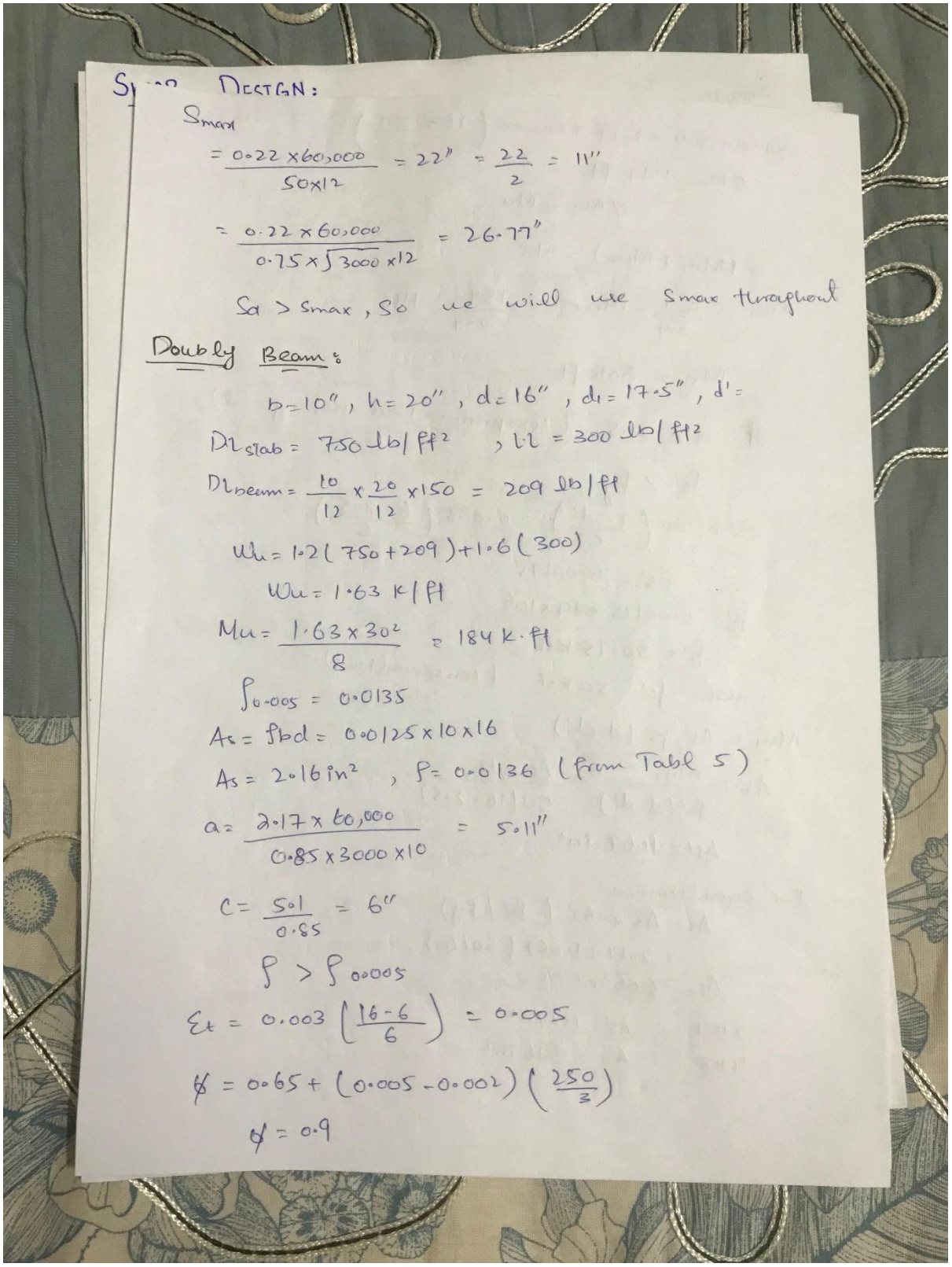
Main bar 2 #10

$d_{agg} = 37.5mm = 1.5''$

- $2''$
- Max $3''$
- $\left(\frac{4}{3}\right) d_{agg} = \frac{4}{3}(1.5)$

$= 2''$ ✓





S... DESIGN:

S_{max}

$= \frac{0.22 \times 60{,}000}{50 \times 12} = 22'' = \frac{22}{2} = 11''$

$= \frac{0.22 \times 60{,}000}{0.75 \times \sqrt{3000} \times 12} = 26.77''$

$S_a > S_{max}$, So we will use S_{max} throughout

Doubly Beam:

$b = 10''$, $h = 20''$, $d = 16''$, $d_1 = 17.5''$, $d' =$

$DL_{slab} = 750\ lb/ft^2$, $LL = 300\ lb/ft^2$

$DL_{beam} = \frac{10}{12} \times \frac{20}{12} \times 150 = 209\ lb/ft$

$W_u = 1.2(750 + 209) + 1.6(300)$

$W_u = 1.63\ K/ft$

$M_u = \frac{1.63 \times 30^2}{8} = 184\ K \cdot ft$

$\rho_{0.005} = 0.0135$

$A_s = \rho bd = 0.0125 \times 10 \times 16$

$A_s = 2.16\ in^2$, $\rho = 0.0136$ (from Table 5)

$a = \frac{2.17 \times 60{,}000}{0.85 \times 3000 \times 10} = 5.11''$

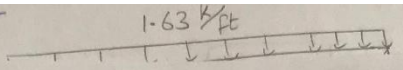
$c = \frac{5.1}{0.85} = 6''$

$\rho > \rho_{0.005}$

$\varepsilon_t = 0.003\left(\frac{16-6}{6}\right) = 0.005$

$\phi = 0.65 + (0.005 - 0.002)\left(\frac{250}{3}\right)$

$\phi = 0.9$



$\phi Mn = 0.9 \times 2.17 \times 60{,}000 \left(16 - \frac{5.1}{2}\right)$

$\phi Mn = 131$ k.ft

$\phi Mn < Mu$

$\phi (Mn_1 + Mn_2) = Mu$

$Mn_1 = \frac{184}{0.9} - Mn_2 = \frac{184}{0.9} - 145$

$Mn_1 = 60$ k.ft

d < 21.5 (compresion bars not yield)

$f_s' < f_y$

$\varepsilon_s' = \varepsilon_u \left(\frac{c - d'}{c}\right) = 0.003 \left(\frac{6 - 2.5}{6}\right)$

$\varepsilon_s' = 0.00175$

$f_s' = 0.00175 \times 29 \times 10^6$

$f_s' = 50.75$ ksi

use $f_s' = 50$ ksi (conservations)

$Mu_1 = As' f_s' (d - d')$

$As' = \frac{Mn_1}{f_s'(d - d')} = \frac{60 \times 12}{50(16 - 2.5)}$

$As' = 1.067$ in²

→ For equilibriom

$As = As + As' (f_s' / f_y)$

$= 2.17 + 1.067 (50/60)$

$As = 3.06$ in²

2#7 . $As' = 1.29$ in²

4#8 $As = 3.16$ in²

SHEAR DESIGN:

→ check ρ

$= 3.16 - 1.2\,(50/60)$

$A_s = 2.16\ in^2$

$$\rho = \frac{A_s}{bd} = \frac{2.16}{10\times16} = 0.0135$$

$$\rho' = \frac{A_s'}{bd} = \frac{1.2}{10\times16} = 0.0075$$

$\rho_{max} = 0.0155$

$\bar{\rho}_{max} = \rho_{max} + \rho' = 0.0155 + 0.0075 = 0.023$

$\bar{\rho}_{0.005} = \rho_{0.005} + \rho' = 0.0135 + 0.0075 = 0.021$

$\rho < \bar{\rho}_{max}$ (steel failure)

$\rho < \bar{\rho}_{0.005}$ ($\phi = 0.9$)

$$\phi M_n = \phi\left[\left[(A_s - A_s'(f_s'/f_y))\, f_y\,(d - a/2)\right] + (A_s' f_s'(d-d')\right]$$

$$a = \frac{(A_s - A_s')\, f_y}{0.85\, f_c' b} = \frac{2.16 \times 60{,}000}{0.85 \times 3000 \times 10}$$

$a = 5.08 \approx 5.1$

$c = 6$

$$\varepsilon_s' = \varepsilon_u\left(\frac{c-d'}{c}\right) = 0.003\left(\frac{6-2.5}{6}\right)$$

$\varepsilon_s' = 0.00175$

$f_s' = 0.00175 \times 29 \times 10^6$

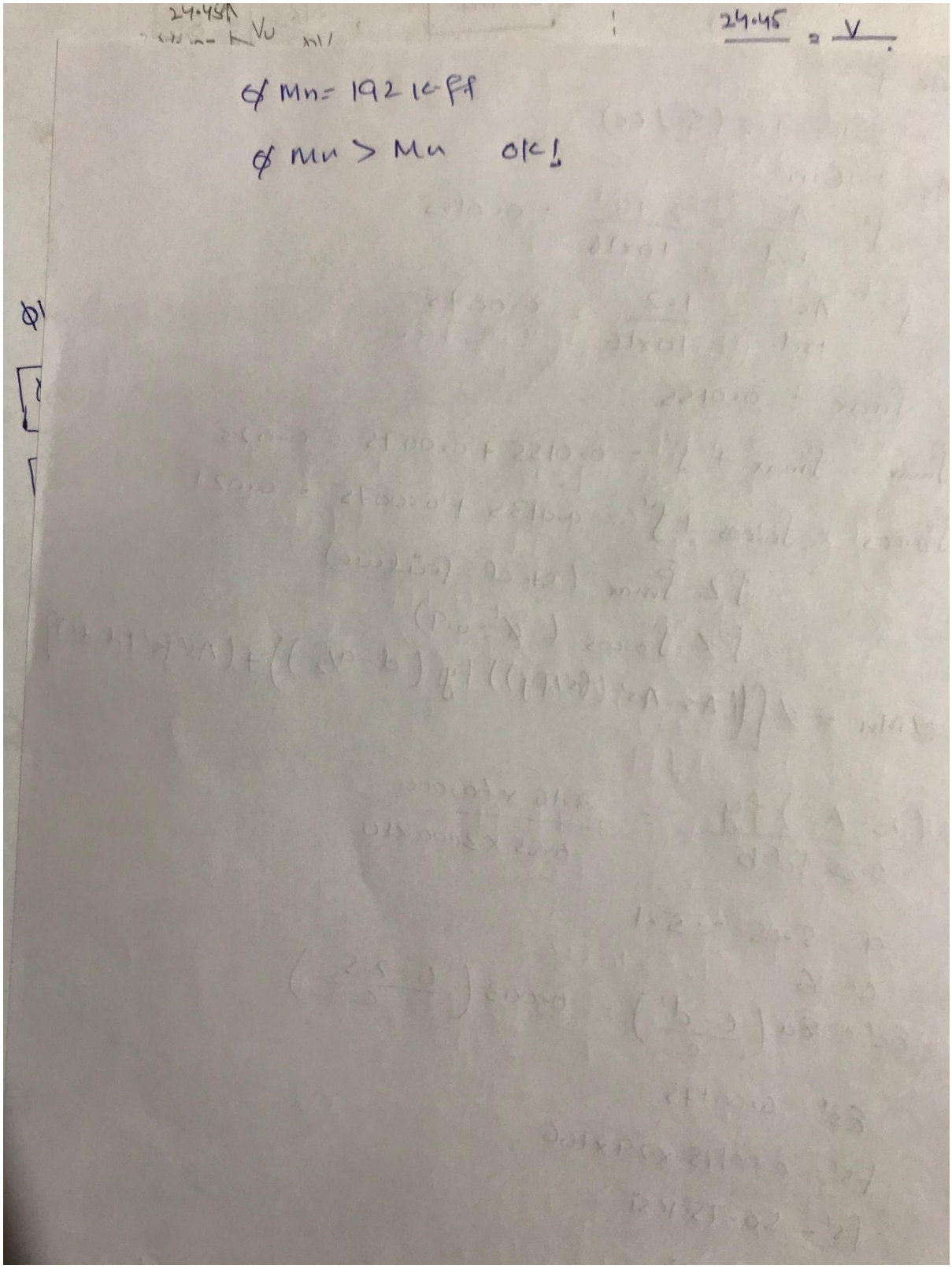
$f_s' = 50.75\ ksi$

$f_s' = 50\ ksi$

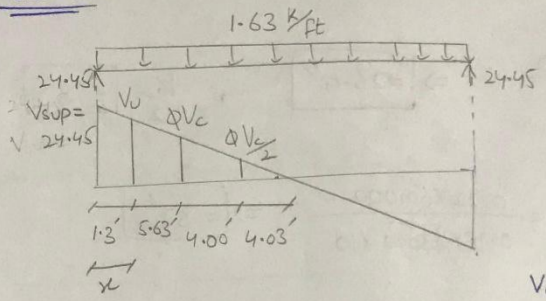
$$\phi M_n = 0.9\left[\left(2.16 \times 60{,}000 + \left(16 - \frac{5.1}{2}\right) + \left(1.2 \times 50\,(16-2.5)\right)\right]$$

$= 0.9\,(145.26 + 67.5)$

$\phi M_n = 212.76 \lesssim 213$


24.45
Vu
24.45 = V
ϕ Mn = 192 k-ft
ϕ Mn > Mu ok!

SHEAR DESIGN:



$$\frac{24.45}{15} = \frac{V}{(15-x)}$$

$$V = 1.63(15-x)$$

$$V_u = 1.63\left(15 - \frac{16}{12}\right)$$

$$\boxed{V_u = 22.3k}$$

$$\Phi V_c = 0.75 \times 2 \times \sqrt{3000} \times 10 \times 16$$

$$\boxed{\Phi V_c = 13.14k}$$

$$\boxed{\frac{\Phi V_c}{2} = 6.57k}$$

$V_u < \Phi V_c$ (web reinforcement required)

$$\Phi V_s = V_u - \Phi V_c$$

$$\Phi V_s = 22.3 - 13.14$$

$$\boxed{V_s = 12.21k}$$

$$V_s < 2V_c$$

$$12.21 < 35.04$$

$$S_d = \frac{\Phi A_v f_{yt} d}{V_u - \Phi V_c}$$

$$= \frac{0.75(0.22)(60{,}000)(16)}{((22.3) - (13.14))(1000)}$$

$S_d = 17.3$ $\boxed{\approx 17''}$

S_{max} :

$= \frac{0.22 \times 60{,}000}{50 \times 10}$ $\Rightarrow \boxed{= 26.4''}$, $\frac{16}{2}$ $\boxed{= 8''}$

$= 24''$, $= \frac{0.22 \times 60{,}000}{0.75 \times \sqrt{3000} \times 10}$ $\Rightarrow \boxed{= 32''}$

$\varepsilon_s = \frac{\varepsilon_{cu}(d-c)}{c} = 29 \times 10^6 (0.003)(6-2)$

DL = 11250 + 4500 + 633 + 1500 = 17883 lbs

$\rightarrow P_u = 1.2(17.883) + 1.6(4.5)$

$P_u = 28.596$ kip $\approx P_u = 29$ kip

- Assume

$e = 0.2b$

$e = 0.2(10'')$

$e = 2$

$M_u = P_u \times e = 29 \times \frac{2}{12} = 4.8$ k.ft ≈ 5 k.ft

Start from 1% steel of gross area

$A_{st} = 0.01(12 \times 12)$

$A_{st} = 1.44$ in^2

4#6

$A_{st} = 1.76$ in^2

cover = 2"

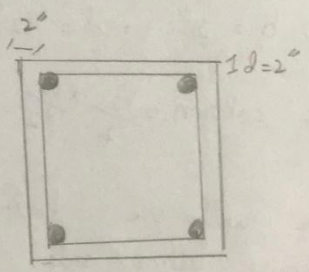
$c = 0.6d$

$c = 0.6(10'')$

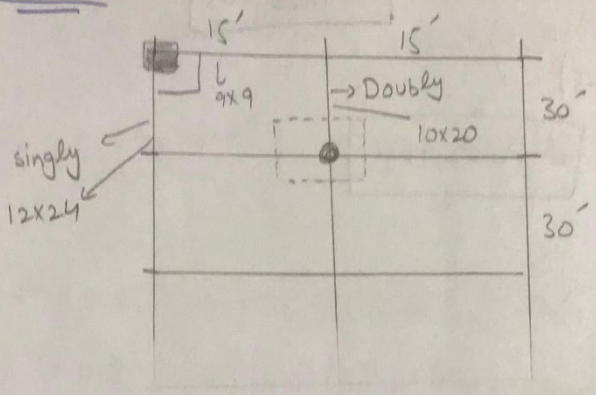
$c = 6$

$a = \beta_1 c = 0.85 \times 6$

$a = 5.1''$



COLUMNS:



→ DL Slab on whole beam = 750 lb/ft × 30 = 22500 lb/2

LL on whole beam = 300 lb/ft × 30 = 9000 lb/2

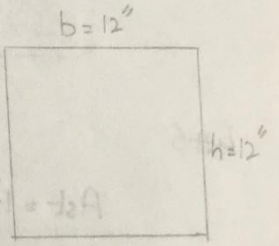
DL singly = 300 lb/ft × 30 = 9000 lb/2

DL doubly = 209 lb/ft × 30 = 6270 lb/2

Rectangular column:

Height of column = 10'

b = 12" , d = 12"



→ DL column = 12/12 × 12/12 × 10 × 150 = 1500 lb

→ We will use half of the loads because other side columns will also bear load.

DL, 9x9 beam = 9/12 × 9/12 × 15 × 150 = 1265.625 ≈ 1266 lb/2

- $f_s' = E_s\left(\frac{\varepsilon_u (c-d')}{c}\right) = 29\times10^6\left(\frac{0.003(6-2)}{6}\right)$ = 58 ksi

- $f_s = E_s\left(\frac{\varepsilon_u (d-c)}{c}\right) = 29\times10^6\left(\frac{0.003(10-6)}{6}\right)$ = 58ksi

→ $P_n = 0.85 f_c' ab + A_s'(f_s' - 0.85 f_c') - A_s f_s$

$= 0.85(3000)(5.1)(12) + \left(\frac{1.76}{2}\right)(58000 - 0.85(3000)) - \left(\frac{1.76}{2}\right)(58000)$

P_n = 153 kip

$\phi P_n = 153 \times 0.65$ = 99kip

→ $M_n = 0.85(3000)(5.1)(12)\left(\frac{12}{2} - \frac{5.1}{2}\right) + \left(\frac{1.76}{2}\right)(58000 - 0.85(3000))\left(\frac{12}{2} - 12^{5}\right)$

$+ \left(\frac{1.76}{2}\right)\times 58000\left(10 - \frac{12}{2}\right)$

$M_n = 538407 + 195184 + 204160$

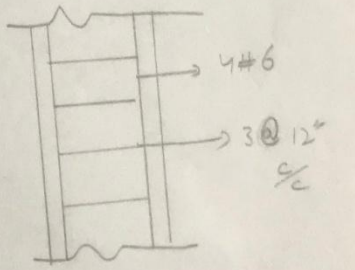
$= \frac{937751}{12000}$

M_n = 78.14 k.ft

$\phi M_n = 0.65 \times 78.14$ ≃ 50 k.ft

$\phi P_n > P_u$

$\phi M_n > M_u$



- $\rho = \frac{A_{st}}{A_g} = \frac{1.76}{12\times12} = 0.012$

= 1.2 %

Beams

Ties:

$16db = 16 \times \frac{6}{8}$ $= 12"$

$48df = 48 \times \frac{3}{8}$ $= 18"$

Smallest dimension $= 12"$

#3@ 12" c/c

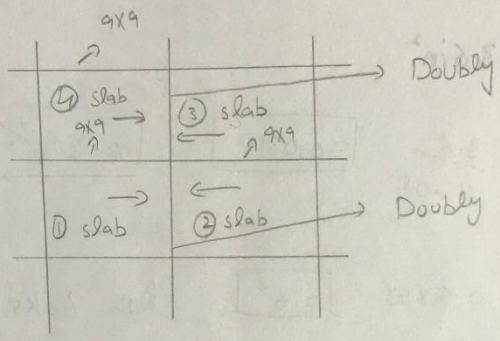
spacing c/c of main bar

2 Doubly Beams = 6270 lb

2(9x9) = 1266 lb

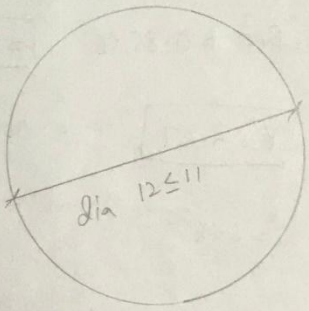
DL slab = 22500 x 2 on (1) whole beam

= 45000 lb



cover = 2"

LL = 9000 x 2 = 18000 lb



Volume = $\frac{\pi}{4} \times d^2 \times h = \frac{\pi}{4} \times 10^2 \times 10$

= 7.85 ~~ft/s~~ x 150 lb ~~/ft/s~~

DL Column = 5890 lb

• $P_U = 1.2(5890 + 6270 + 1266 + 45000) + 1.6(18000)$

$P_U = 98.9$ ≈ 99 kip

$\ldots\ldots 11 \times 12 - 5.1\Big) + \left(\frac{3.6}{2}\right)(58000 - 0.85(3000))\left(\frac{12}{2} - 2\right)$

$l = 0.2d = 0.2 \times 10 = 2$

$M_u = P_u \times l$

$= 99 \times \frac{2}{12}$

$= 16.5 \text{ k.ft}$

$\rho = 0.03 = 3\%$

$A_g = \frac{\pi}{4} d^2 = \frac{\pi}{4}(12)^2$

$= 113.01 \text{ in}^2 \times 0.03$

$A_{st} = 3.4 \text{ in}^2$

6#7 $A_{st} = 3.6 \text{ in}^2$

- $\rho = \frac{A_{st}}{A_g} = \frac{3.6}{113.01} = 0.0318 \simeq 3.18\%$

$c = 0.6d = 0.6 \times 10 = 6''$ for 60ksi = fy

$a = \beta_1 c = 0.85 \times 6 = 5.1''$

$\gamma = 0.67$

$\rho = 3.18\%$

- $f_s = 29 \times 10^6 \left(\frac{0.003(10-6)}{6}\right) = 58 \text{ ksi}$

$f_s' = 29 \times 10^6 \left(\frac{0.003(6-2)}{6}\right) = 58 \text{ ksi}$

$\rightarrow P_n = 0.85 \times 3000 \times 5.1 \times 12 + \left(\frac{3.6}{2}\right)(58000 - 0.85(3000)) - \left(\frac{3.6}{2}\right)(58000)$

$P_n = 151 \text{ kip}$

3(8) = 24" , So ok!

Main reinforcement #3 @ 8" c/c

$A_{s\min} = 0.0018\,bh = 0.0018 \times 12 \times 8$

$A_{s\min} = 0.17\ in^2/ft$ OK!

#3 @ 8" c/c (shrinkage)

5(h) = 5(8) = 40" , 18" So spacing ok!

=> checking shear:

$$V_u = \frac{w_u l}{2} = \frac{0.184 \times 15}{2}$$

$$\boxed{V_u = 1.38\ K}$$

$$\phi V_c = 0.75 \times 2 \sqrt{f_c'}\, bd$$

$$= 0.75 \times 2 \sqrt{3000} \times 12 \times 7$$

$$\phi V_c = 6.9\ K$$

$$\phi V_c > V_u$$

No shear reinforcement required

BEAM:

→ Slab load

D.L = 100 lb/ft²/2 , LL = 40 lb/ft²/2

D.L = 50 lb/ft² , LL = 20 lb/ft²

→ One beam bear this load

D.L = 50 × 15 = 750 , LL = 20 × 15 = 300

DL = 3.33 lb/ft² , LL = 1.33 lb/ft

→ $h_{beam} = \frac{30 \times 12}{18.5} = 19.45" \approx 20"$

cover = 2"

$\rightarrow M_U = 0.85(3000)(5.1)(12)\left(\frac{12}{2} - \frac{5.1}{2}\right) + \left(\frac{3.6}{2}\right)(58000 - 0.85(3000))\left(\frac{12}{2} - 2\right)$

$+ \left(\frac{3.6}{2}\right)(58000)\left(10 - \frac{12}{2}\right)$

$M_U = \frac{538407 + 399240 + 417600}{12000}$

$M_U = 113\,K\text{-}ft$

$\phi P_n = 0.75 \times 151 \quad = 113\,kip$

$\phi M_n = 0.75 \times 113 \quad = 84\,K\text{-}ft$

$\rho_{min} < \rho < \rho_{max}$

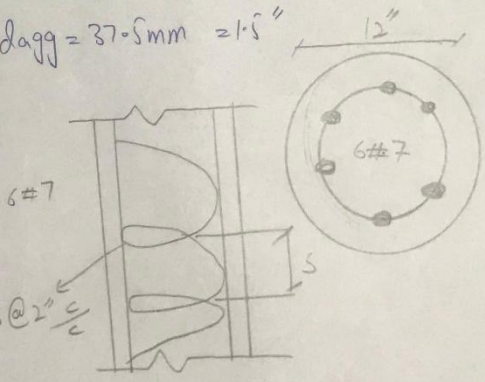
$\phi P_n > P_u$

$\phi M_n > M_U$ ok

Ties:

Main bar 2 #10

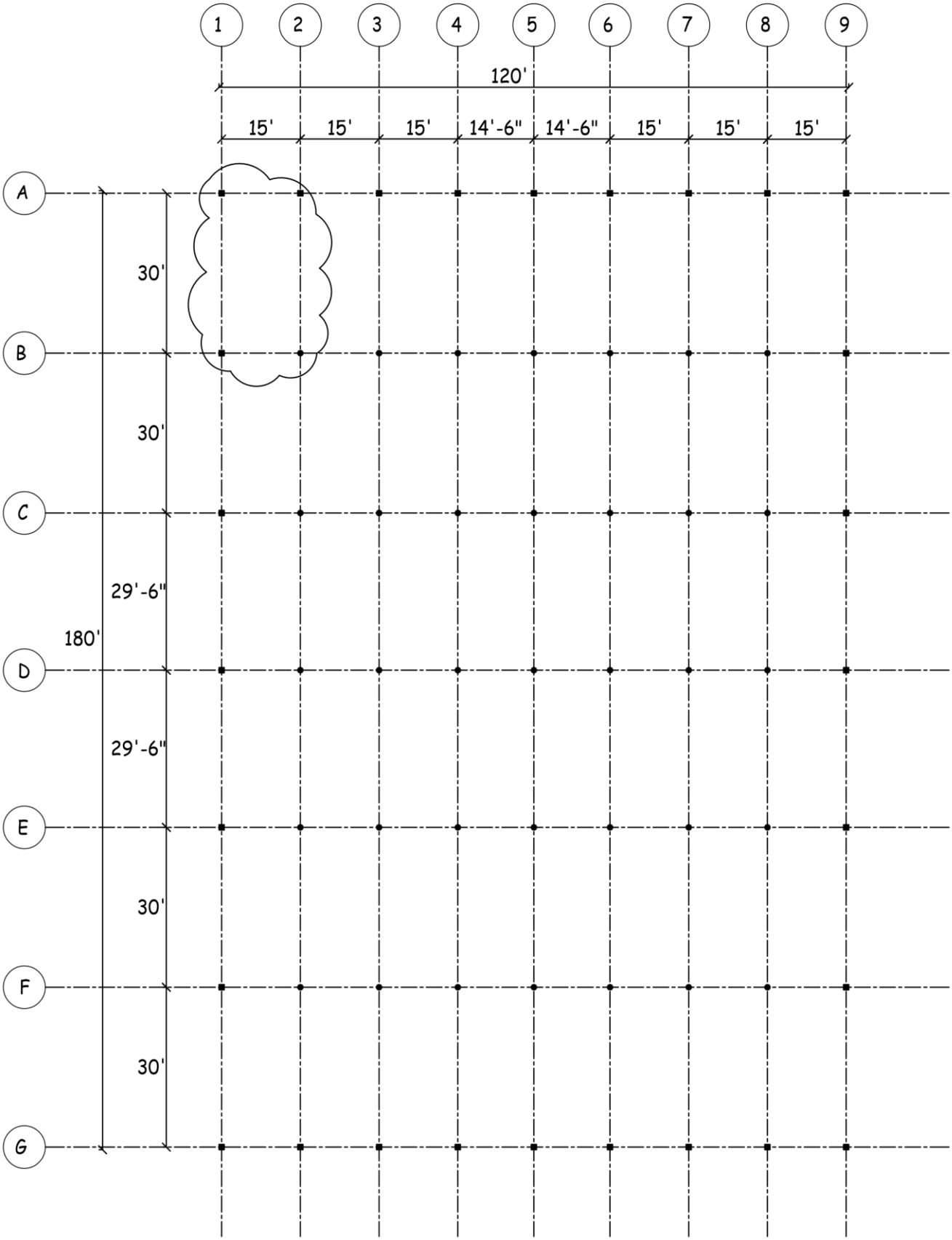
$d_{agg} = 37.5mm = 1.5''$



- 2"
- Max 3"
- $\left(\frac{4}{3}\right) d_{agg} = \frac{4}{3}(1.5)$

$= 2''$ ✓

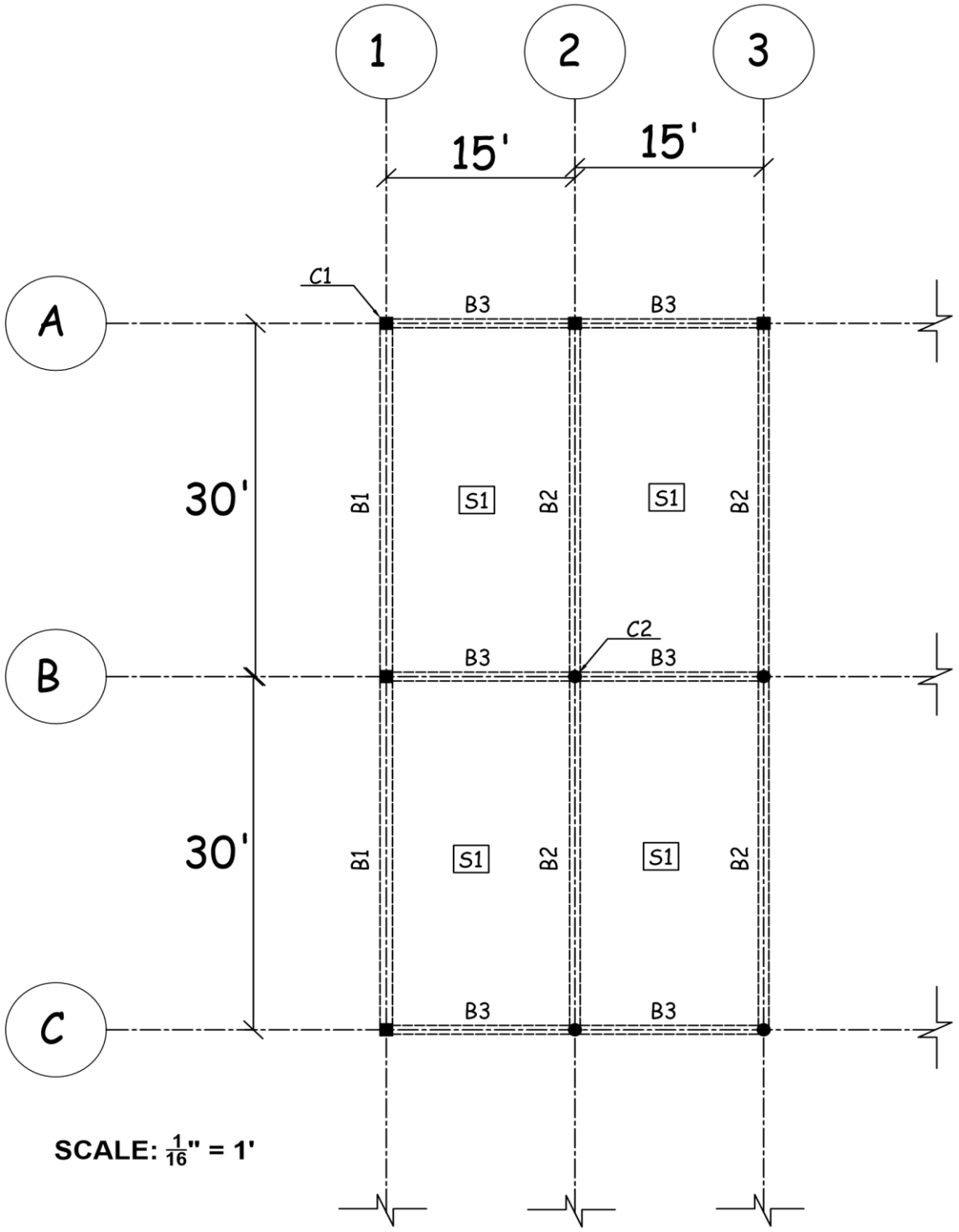
GRID LAYOUT



SCALE: $\frac{1}{24}$" = 1'

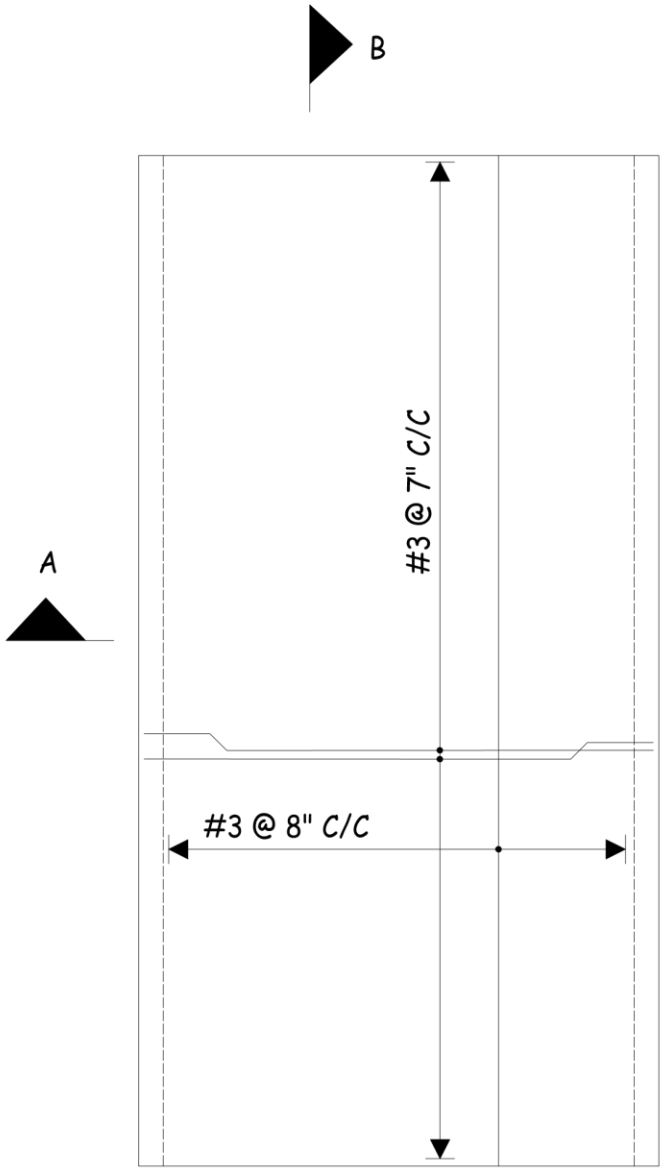
REPRESENTS AREA THAT IS STUDIED

DESIGNED COLUMN BEAM AND SLAB

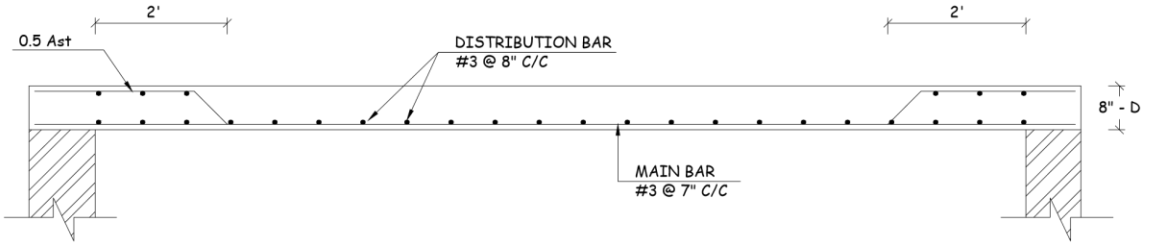


	C1	C2	B1	B2	B3	S1
DIMENSIONS	12" x 12"	12" Ø	12" x 24"	10" x 20"	9" x 9"	8" THICK ONE WAY SLAB

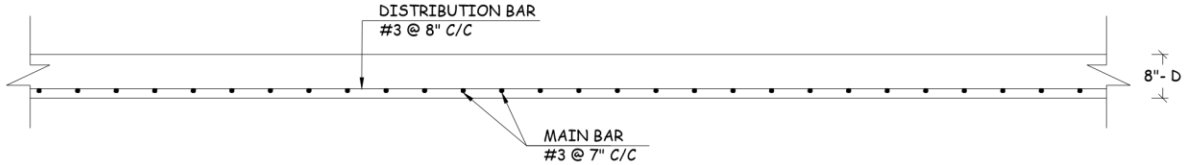
SLAB



PLAN
SCALE: $\frac{1}{8}$"=1'

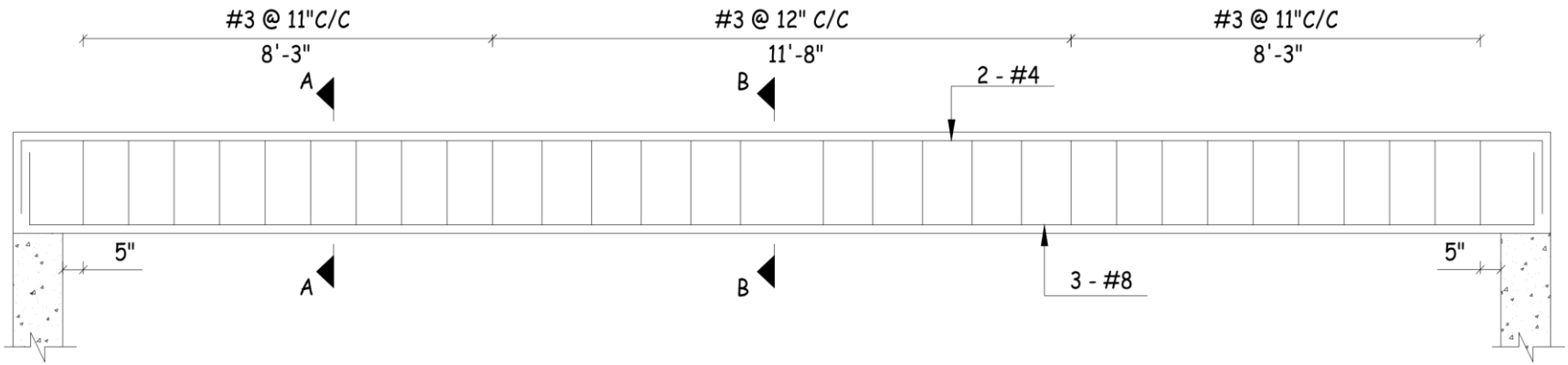


SECTION A-A
SCALE: $\frac{3}{8}$"=1'

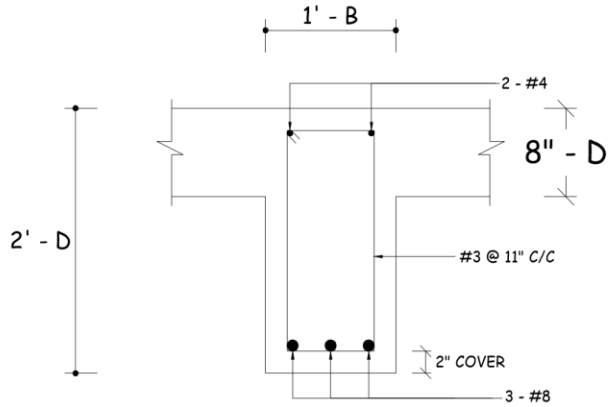


SECTION B-B
SCALE: $\frac{3}{8}$"=1'

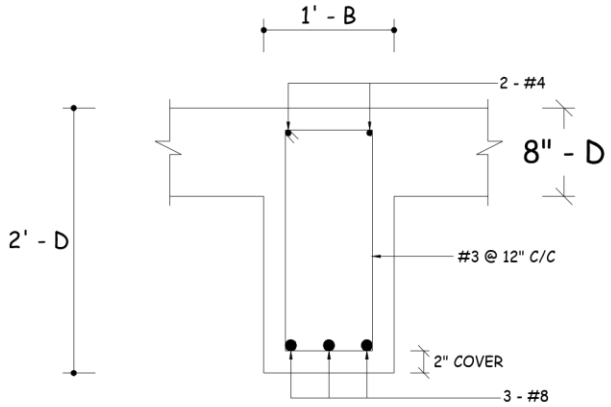
BEAM DETAILING



B1 LONG. SECTION
SCALE: $\frac{5}{16}$"=1'

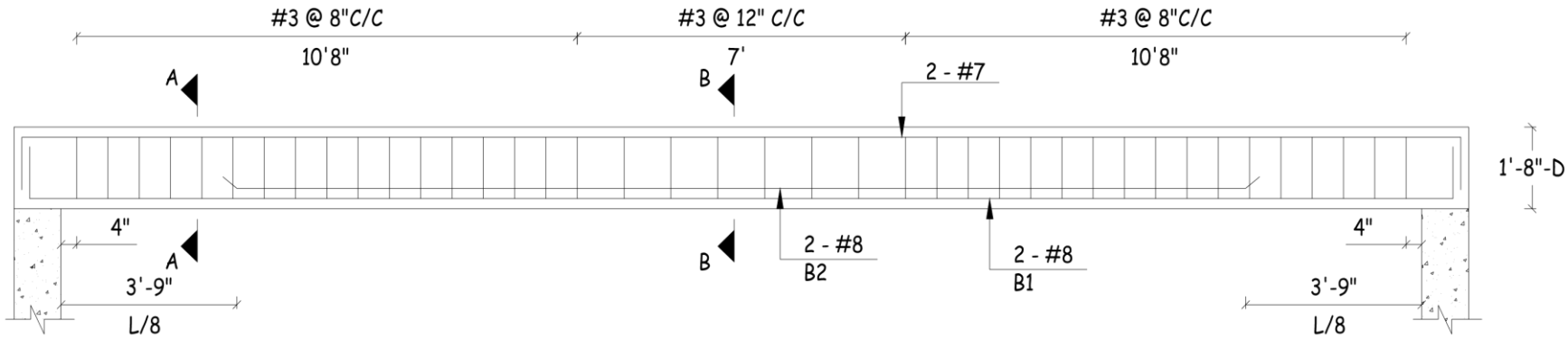


SECTION A-A
SCALE: $\frac{3}{4}$"=1'

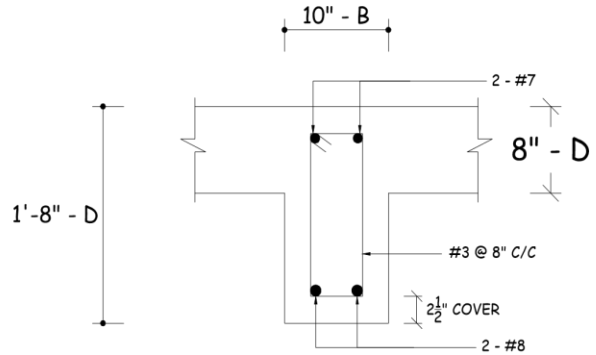


SECTION B-B
SCALE: $\frac{3}{4}$"=1'

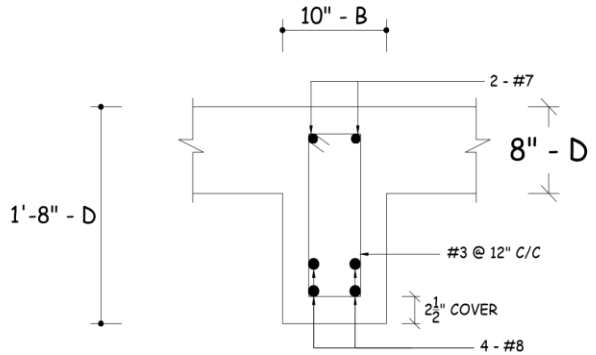
BEAM DETAILING



B2 LONG. SECTION
SCALE: $\frac{5}{16}$"=1'

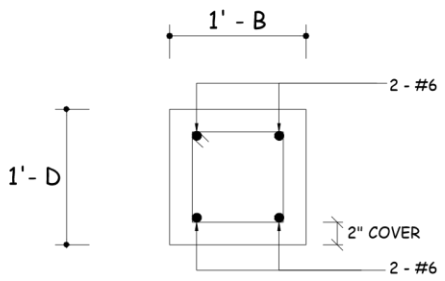


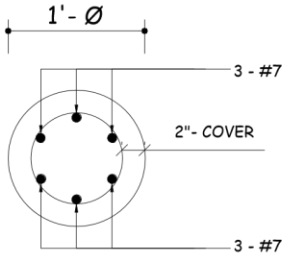
SECTION A-A
SCALE: $\frac{3}{4}$"=1'

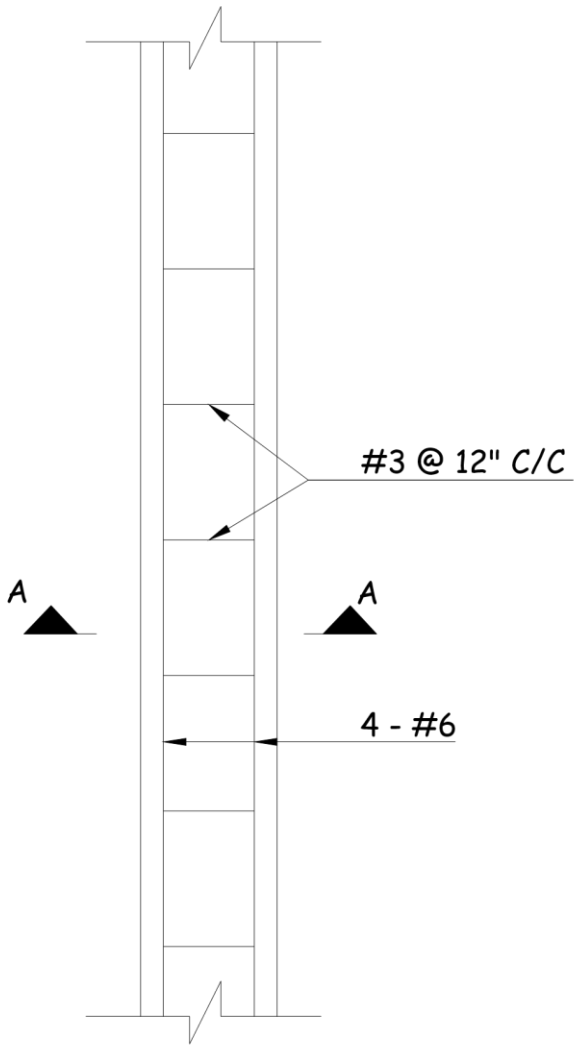


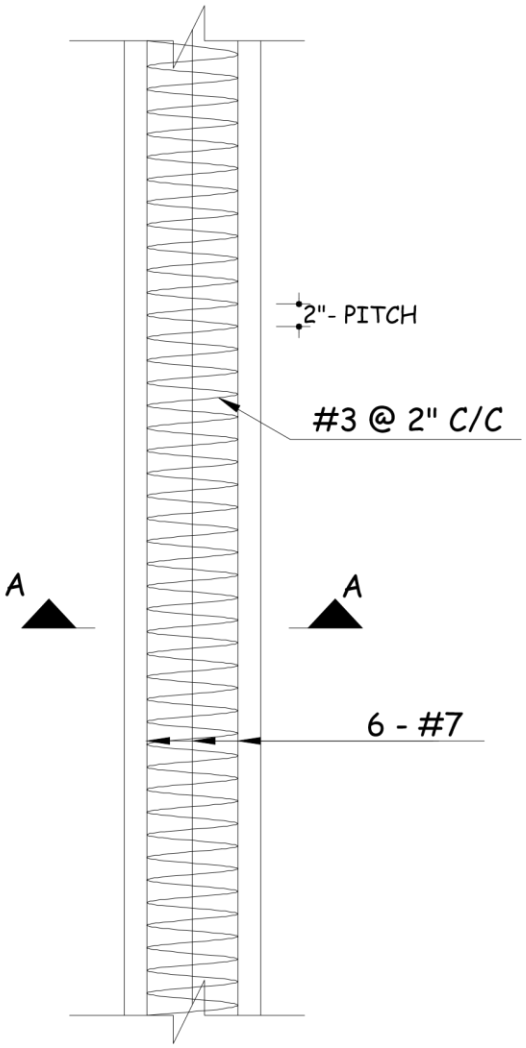
SECTION B-B
SCALE: $\frac{3}{4}$"=1'

COLUMN DETAILING









5. Design Calculations and Details

5.1 One-Way Slab

The one-way slab was designed with a thickness of 8 inches to satisfy deflection control requirements based on ACI Tables 7.3.1.1 and 9.3.1.1. The main reinforcement provided is **#3 @ 7" c/c** while distribution reinforcement consists of **#3 @ 8" c/c.** To enhance anchorage and reduce negative moments at supports, a 2-foot crank length was provided at both ends of the slab. The slab rests on support with a width of 12 inches, and the reinforcement configuration ensures structural efficiency and serviceability.

5.2 Singly Reinforced Beam (B1)

Beam B1 is a simply supported beam with dimensions of 1' in width and 2' in depth. It is reinforced with three #8 bars at the bottom as the main tensile reinforcement. Shear resistance is provided by two-legged **#3 @ 11" c/c** for the first 8' 3" from both supports, while at the mid-span, stirrups are spaced at **12" c/c**. A clear cover of **2"** is maintained throughout to ensure durability and protection of reinforcement.

5.3 Doubly Reinforced Beam (B2)

The doubly reinforced beam, B2, has a cross-section of 10" wide and 20" deep. The bottom reinforcement comprises four #8 bars, with two of them curtailed 3' 9" away from each support, satisfying ACI bar cut-off requirements. At the top, two #7 bars are placed to resist compressive stresses. For shear reinforcement, #3 @ 8" c/c up to 10'8" from supports, transitioning to 12" spacing in the central 7' region. A clear cover of 2" is used to maintain corrosion resistance.

5.4 Exterior Square Column (C1)

Column C1 is a square column with dimensions of 12" by 12". It is reinforced longitudinally with four #6 bars placed symmetrically, providing adequate axial load resistance. Lateral ties

are provided using #3 @ 12” c/c. A 2-inch clear cover is maintained to meet durability requirements and ensure proper confinement of reinforcement.

5.5 Circular Spiral Column (C2)

The circular interior column, C2, has a diameter of 12” and is reinforced with six #7 longitudinal bars arranged evenly along the perimeter. Spiral confinement is provided using #3 bars wound at a 2” pitch, ensuring adequate ductility and load resistance. A clear cover of 2” is maintained, and the design conforms to ACI provisions for spiral reinforcement and confinement in compression members

6. Comparison and Discussion

Slabs							
	Group 3	**Group 9**	**Group 2 or 8**	**Group 1 or 7**	**Group 4 or 10**	**Group 5 or 11**	**Group 6 or 12**
Occupancy	**School/College**	School/College	Office	Residential	**Library**	Gymnasium	Hotel
Span (ft)	15		15	15	14		20
Depth (inch)	8		10	8	7”		11
Area (sq. ft)	450		450	450	392		240
Support condition	One end continuous		One end continuous	One end continuous	One end continuous		One end continuous
Live load (psf)	40		100	40	60		85
Dead load (psf)	100		125	100	87.5l		530
Wu (k/ft)	0.184		0.31	0.184	0.201		0.772
Mu (k-ft)	5.175		8.72	5.175	4.925		38.6
Vu (k)	1.38		2.325	1.38	1.407		
ρ(req)	0.002		0.0026	0.002	0.0026		0.0018
As(provided)	0.18		0.2	0.18	0.187		0.2592
ϕMn(k-ft)	5.175		7.025	5.175	4.925		
Negative ref bars							
Positive ref bars	#3@7”c/c		#4@9” c/c	#3@7.5”c/c	#3 @ 7” c/c		#6 @ 6” c/c
Shear ref bars (if req)	No shear ref bars required				No shear ref bars required		
Δ(LL)	Under Control		0		No		0
Distribution bars	#3@8”c/c		#3@6” c/c	#3@7”c/c	#3 @ 9” c/c		#3 @ 10” c/c
Dev. length of positive bars	4.62”		8.2”	9.24”	6.93”		6.5”
Dev. length of negative bars							
Spacing as per ACI cracking control criteria	Under Control		Under Control	Under Control	Under Control		Under Control

Single reinforced beam							
	Group 3	**Group 9**	**Group 2 or 8**	**Group 1 or 7**	**Group 4 or 10**	**Group 5 or 11**	**Group 6 or 12**
Occupancy	**School/College**	School/College	Office	Residential	**Library**	Gymnasium	Hotel
Span (ft)	30		30	30	28		30
Support condition	One end continuous		One end continuous	One end continuous	One end continuous		One end continuous
Depth (inch)	24		26	26	21		24
Width (inch)	12		12	12	12		12
Dead load (k/ft)	1.05		1.2625	1.253	0.263		0.338
Wu (k/ft)	1.74		2.715	1.734	1.7232		4.992
Mu (k-ft)	196		305.44	195.07	168.87		249.6
Vu (k)	23		35.295	26.01	21.48		40.98
ρ(req) for flexure	0.0089		0.0113	0.0084	0.0098		0.0116
ρ(req) for shear	0.00167		0.00153	0.0031	0.00203		0.0004
Under or over ref?	Under		Under reinforced	Under	Under ref		Under ref
ϕMn(k-ft)	210		347.4	225.65	178		250.6
ϕVn(kips)	28.5		35.29	23.17	21.48		41.9
Negative ref bars	2#4 bars		2 #8 bars	2 #4 bars	2 #4 bars		
Positive ref bars	3#8 bars		3#10 bars	3#8 bars	3 #8 bars		3 #9 bars
Shear ref bars (if req)	2 legged #3 stirrup @11”		2 legged #3 stirrup @ 12” c/c	2 legged #3 stirrup @ 6” c/c	2 legged #3 stirrup @ 9” c/c		2 legged #3 stirrup @ 10” c/c
Δ(LL)	Under Control		0		0		0
Dev. length of positive bars	41”		66”	82.158”	41”		32.86”
Dev. length of negative bars	22”		41”	16.431”	10.27”		
Spacing as per ACI cracking control criteria	Under Control		Under Control	Under Control	Under Control		Under Control
Cross-section criteria satisfied?	Yes		yes	Yes	Yes		Yes

Doubly reinforced beam							
	Group 3	**Group 9**	**Group 2 or 8**	**Group 1 or 7**	**Group 4 or 10**	**Group 5 or 11**	**Group 6 or 12**
Occupancy	**School/College**	School/College	Office	Residential	**Library**	Gymnasium	Hotel
Span (ft)	30		30	30	28		30”
Support condition	One end continuous		One end continuous	One end continuous	One end continuous		One end continuous
Depth (inch)	20		20	20	20		22
Width (inch)	10		10	12	10		12
Dead load (k/ft)	0.959		1.1458	1.14	0.613		3.45
Wu (k/ft)	1.63		2.57	1.62	1.66		4.9
Mu (k-ft)	184		289	182.25	162.17		248.1
Vu (k)	22.3		53.13	24.3	20.6		43.22
ρ(req) for flexure	0.0135		0.0155	0.0135	0.0155		0.0135
ρ’(req)	0.0075		0.0195	0.0045	0.0038		0.0005
Comp bars yielding (Yes/No)?	No		No	No	No		No
ρ(req) for shear	0.00275		0.00314	0.0023	0.00229		0.0005
Under or over ref?	Under		Under reinforced	Under ref	Under ref		Under ref
ϕMn(k-ft)	192		352.8	203.35	173		210
ϕVn(kips)	22.3		45.62	22.145	20.6		33.46
Negative ref bars	2#7 bars		2#11 bars	2#6 bars	2 #5 bars		1 #3 bars
Positive ref bars	4#8 bars		4#10 bars	4#9 bars	5 #7 bars		5 #7 bars
Shear ref bars (if req)	2 legged #3 stirrup @11”c/c		2 legged #3 stirrup @4”c/c	2 legged #3 stirrup @8”	2 legged #3 stirrup @9”c/c		2 legged #3 stirrup @10”c/c
Δ(LL)	Under Control		0	0	0		0
Dev. length of positive bars	33”		33.13”	104”	25”		25”
Dev. length of negative bars	26”		65.3”	46.21”	13”		3.7”
Spacing as per ACI cracking control criteria	Under Control		Under Control	Under Control	Under Control		Under Control
Cross-section criteria satisfied?	Yes		Yes	Yes	Yes		Yes

Interior Spiral Column							
	Group 3	**Group 9**	**Group 2 or 8**	**Group 1 or 7**	**Group 4 or 10**	**Group 5 or 11**	**Group 6 or 12**
Occupancy	**School/College**	School/College	Office	Residential	**Library**	Gymnasium	Hotel
Dia. (inch)	12		15	18	12		16
Height (ft)	10		10	11	11		10
Concentric or Eccentric Col.?	Eccentric		Eccentric	Eccentric	Eccentric		Eccentric
e (inch)	2		2.6	3.6	2		2.8
e/D	0.2		0.17	0.2	0.2		0.28
Dead load (kip)	54		65.6	62.4	50		135.03
Pu (kip)	99		150.73	158	109		194
Mu (k-ft)	16.5		32.66	47.05	18.17		45
ρ(req)	0.0318		0.0246	0.02	0.03		0.029
ϕPn(kip)	113		184.9	320	113.6		276
ϕMn(k-ft)	84		178	205	85.8		56
Main ref bars	6#7		7#8 bars	6#9 bars	6 #7 bars		6#9 bars
Lateral ref bars	#3@2”c/c		#3@2” c/c	#3@2” c/c	#3 @ 2”c/c		#3 @ 2”c/c
Spacing criteria satisfied for both ref.?	Yes		Yes	Yes	Yes		Yes

Exterior Square Column							
	Group 3	**Group 9**	**Group 2 or 8**	**Group 1 or 7**	**Group 4 or 10**	**Group 5 or 11**	**Group 6 or 12**
Occupancy	**School/College**	School/College	Office	Residential	**Library**	Gymnasium	Hotel
Width (inch)	12		12	16	12		14
Height (ft)	10		10	11	11		10
Concentric or Eccentric Col.?	Eccentric		Eccentric	Eccentric	Eccentric		Eccentric
e (inch)	2		2	3	2.4		2.4
e/D	0.2		0.2	0.18	0.2		0.24
Dead load (kip)	18		21.0705	19.9	14		37
Pu (kips)	29		43.28	44.85	33		52
Mu (k-ft)	5		7.2	11.21	8.42		10
ρ(req)	0.012		0.246	0.01	0.01		0.012
ϕPn(kip)	99		98.87	171.13	33.7		138
ϕMn(k-ft)	78		50.9	117	8.5		45
Main ref bars	4#6 bars		4#6 bars	8#6 bars	4 #6 bars		10 #4 bars
Lateral ref bars	#3@12”c/c		#3@12” c/c	#3@12” c/c	#3 @ 12” c/c		#3 @ 8” c/c
Spacing criteria satisfied for both ref.?	Yes		yes	Yes	Yes		Yes

7. Discussion

- Comparison is between the Occupancy of school and library

One-Way Slab

- **Span & Area:**
  - Group 3: 15 ft span, 450 sq. ft area.
  - Group 4: 14 ft span, 392 sq. ft area.
  - *Slightly shorter span in the library; however, higher loading leads to increased design demand.*
- **Loading:**
  - Live Load: Group 3 = 40 psf; Group 4 = 60 psf.
  - Dead Load: Group 3 = 100 psf; Group 4 = 87.5 psf.
  - *Library slab experiences higher live loads due to public use and shelving systems.*
- **Moment & Reinforcement:**
  - Group 3 has slightly higher Mu (5.175 k-ft), but Group 4 has higher ρ (0.0026 vs 0.002), indicating higher reinforcement need per unit width.
  - Development length is longer in Group 4 (6.93” vs 4.62”) due to higher stress.
- **Conclusion:**
  - *Library slabs are more reinforcement-intensive to cater for dynamic use and heavier occupancy loads.*

Singly Reinforced Beam

- **Span & Dimensions:**
  - Group 3: 30 ft span, 24” depth.
  - Group 4: 28 ft span, 21” depth.

- *Slightly shorter span in Group 4 but still carries significant load due to occupancy.*

- **Loading & Strength:**
  - Mu: Group 3 = 196 k-ft; Group 4 = 168.87 k-ft.
  - ρ(flexure): Slightly higher in Group 4, showing a need for more reinforcement despite lower Mu.
- **Shear Bars:**
  - Closer stirrup spacing in Group 4 (#3 @ 9" c/c vs 11") to resist higher shear demand.
- **Conclusion:**
  - *Library beams are slightly more conservatively designed in shear due to risk of sudden failure and denser usage.*

Doubly Reinforced Beam

- **Flexure & Reinforcement:**
  - Both groups have equal span (30 ft) and same section size.
  - Group 4 has more positive bars (5#7 vs 4#8), indicating slightly greater flexural demand.
  - ρ' (compression reinforcement): Group 3 uses more (0.0075 vs 0.0038), showing a balance in compression/tension steel.
- **Development Lengths:**
  - Group 3: Longer development lengths (33" vs 25") suggest higher stress levels and anchorage requirements.

- **Conclusion:**
    - *Group 4 beams optimize steel use with more efficient rebar selection and spacing to accommodate load variations in library spaces.*

Interior Spiral Column

- **Loads & Size:**
    - Both groups use same diameter (12”), but Group 4 has slightly taller columns (11 ft vs 10 ft).
    - Axial Load Pu: Group 4 = 109 kips; Group 3 = 99 kips.
    - Mu: Slightly higher in Group 4 (18.17 k-ft vs 16.5 k-ft).
- **Reinforcement:**
    - Both groups use same main bars (6#7) and spiral ties (#3 @ 2” c/c), indicating similar confinement strategies.
- **Conclusion:**
    - *Variation in loads is small but important. Library columns are designed for slightly higher eccentric moments and height, contributing to stability in open-floor library layouts.*

Exterior Square Column

- **Height & Load:**
    - Group 3: 10 ft high; Group 4: 11 ft high.
    - Group 4 has higher Pu and Mu, indicating higher axial and moment demands.
- **Moment & Reinforcement:**
    - Moment in Group 4 is 8.42 k-ft (vs 5 k-ft in Group 3), likely due to lateral loading or asymmetrical library layouts.

- Both groups use the same main bars (4#6), but Group 4 has slightly higher development length need due to longer column.

- **Conclusion:**
  - *Exterior library columns handle higher lateral and gravity loads; hence greater care in confinement and stiffness is required.*

Why Variations Exist

1. **Occupancy Function:**
   - Libraries need to support dense shelving, higher crowd loads, and longer durations of occupancy per square foot.
   - Schools/colleges have lighter and more uniformly distributed loads.
2. **Live Load Design Standards (per ACI):**
   - Libraries typically require higher design live loads (60 psf or more) than classrooms (40 psf).
3. **Usage Patterns:**
   - Library structures experience concentrated point loads (bookshelves) and vibration-sensitive usage.
   - Classrooms have more distributed loads and lower vibration sensitivity.
4. **Column Height Differences:**
   - Group 4 columns are taller (11 ft vs 10 ft), increasing the slenderness ratio and requiring more moment resistance.
5. **Serviceability & Deflection:**
   - Library members need tighter control over deflection and cracking due to long spans and continuous occupation.

6. **Safety & Redundancy:**
    - Library structures may be designed with greater safety factors due to public access and load unpredictability.

General Observations on Other Groups

- **Residential (Group 1/7)**: Lightest design overall — smallest loads and reinforcement, suitable for homes.
- **Office (Group 2/8)**: Moderate loads, design lies between residential and school.
- **Gymnasium (Group 5/11)**: Like Library in some cases, very high dead loads (possibly due to equipment).
- **Hotel (Group 6/12)**: Heavily loaded but mainly focused on consistent live/dead load balance.

8. Conclusion

This CEP project provided valuable insight into the structural design of key reinforced concrete members—slabs, beams, and columns—tailored to specific building occupancies. By comparing our school/college-based design (Group 3) with that of a library (Group 4), it became evident that occupancy type significantly impacts structural requirements, with libraries demanding larger sections and higher reinforcement due to greater live and dead loads. The exercise highlighted the importance of aligning design decisions with functional use, adhering to ACI 318-14 standards, and balancing strength, serviceability, and efficiency in practical applications.

9. References

1. ACI 318-14: Building Code Requirements for Structural Concrete
2. ASCE 7: Minimum Design Loads for Buildings and Other Structures
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